

Faithful Acoustic Solver for Extended Spaces

User Manual

Version 0.9

Vito Romanelli Tricanico

October 18, 2026

Contents

1 Introduction **3**

2 Installation **5**
- 2.1 System requirements . . . 5
- 2.2 Installing with `install.sh` . . . 5
- 2.3 Starting F.A.S.E.S . . . 6
- 2.4 Updating . . . 6
- 2.5 Uninstalling . . . 6
- 2.6 Windows: running F.A.S.E.S in WSL . . . 7
- 2.7 Running from the source code (developers) . . . 8

3 Quick start **10**

4 How F.A.S.E.S simulates **11**
- 4.1 The hybrid scheme and the crossover frequency . . . 11
- 4.2 The wave-based solver (FVM) . . . 11
- 4.3 Geometrical acoustics (GA) . . . 12
- 4.4 Combining the parts . . . 13
- 4.5 Units and coordinates . . . 13

5 The main window **14**
- 5.1 Navigating the 3D view . . . 14
- 5.2 The scene menu (right click on 3D view) . . . 15

6 Scene and materials **16**
- 6.1 Projects . . . 16
- 6.2 Importing the scene . . . 16
- 6.3 The material table . . . 16

7 Sources and receivers **19**
- 7.1 The point tables . . . 19
- 7.2 Placing points in the 3D view . . . 20
- 7.3 Source settings . . . 21
- 7.4 Receiver settings . . . 21

8 Setup **23**
- 8.1 Fields . . . 24
- 8.2 PML faces . . . 24
- 8.3 Saving the setup . . . 24
- 8.4 Choosing the values . . . 24

9 Running simulations **26**
9.1 Simulation . . . 27
9.2 The Output log . . . 27
9.3 Ray-path overlay . . . 27
9.4 Animation . . . 28
9.5 Modes . . . 28

10 Results **30**
10.1 Views . . . 30
10.2 Zoom, save and export . . . 31
10.3 Comparing simulations . . . 32

11 Auralization **33**
11.1 Head pose . . . 33
11.2 Binaural rendering (Convolve) . . . 33
11.3 Live ambisonic playback . . . 33
11.4 Receiver view . . . 34
11.5 Per-source signals . . . 34

12 Draw mode: sketcher and props **35**
12.1 The 2D sketcher . . . 35
12.2 Props . . . 36

13 Projects and files **38**
13.1 The project folder . . . 38
13.2 `config.txt` . . . 38
13.3 Sources, receivers and materials . . . 39
13.4 Output files . . . 39

14 Tips and troubleshooting **40**
14.1 Accuracy . . . 40
14.2 Common problems . . . 40
14.3 Getting help . . . 41

A Command-line reference **42**
A.1 The solver . . . 42
A.2 The Python tools . . . 42

B Licence and acknowledgements **43**
B.1 Licence . . . 43
B.2 Third-party software . . . 43
B.3 Citing F.A.S.E.S . . . 43

C Miscelaneous Results **44**

Chapter 1

Introduction

F.A.S.E.S is an acoustic simulation environment for enclosed and open spaces: rooms, halls, churches and warehouses, but also stages, open-air theatres, courtyards, factories and small urban spaces such as squares and streets. It predicts the sound field of a space from its 3D geometry and its materials, computes impulse responses and the ISO 3382 acoustic parameters, lets you listen to the result (auralization), and visualizes how sound propagates and which modes form. Open sides are modelled with absorbing boundaries (Section 8.2), so a space does not need to be watertight.

F.A.S.E.S is a *hybrid* simulator: it combines two families of methods, each used where it is accurate.

- **Finite Volume Method (FVM)** for the frequencies below a chosen *cutoff frequency*. A finite-volume time-domain solver running on a GPU solves the acoustic wave equation on a 3D grid. It captures diffraction, interference and modes exactly, including frequency-dependent surface absorption.
- **Geometrical acoustics (GA)** for the higher frequencies, where the wavelength is small compared to the space: an image-source method (ISM) for the early specular reflections and a ray tracer with scattering for the reverberant tail.

The two are later joined at a *crossover frequency* into one broadband impulse response.

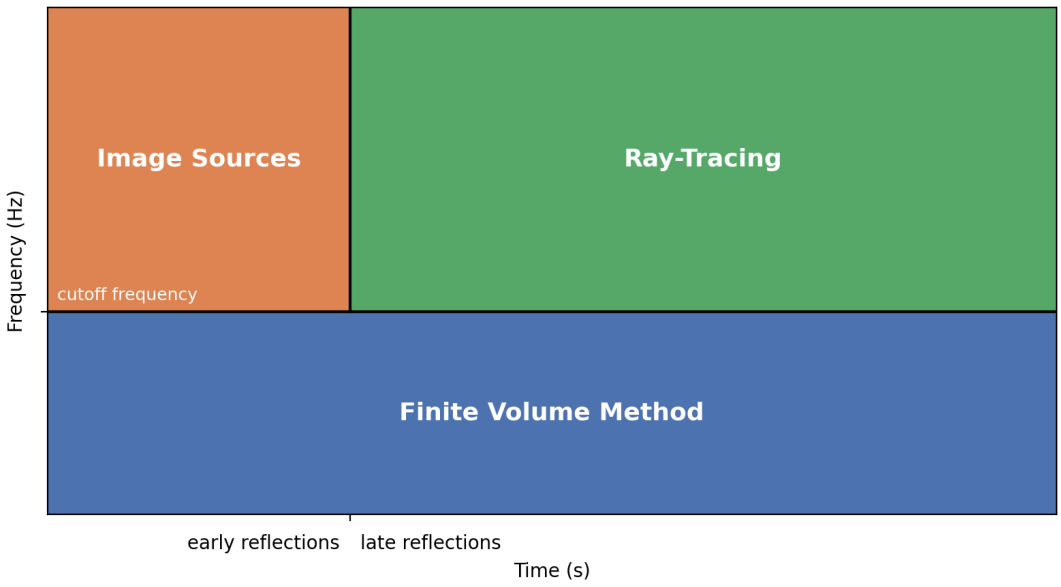


Figure 1.1: How each method is used in F.A.S.E.S in function of frequency and time

Main features

- 3D scene import (OBJ) with one material per layer; a 2D sketcher to draw spaces from scratch or from a DXF floor plan; placeable 3D props (seats, stairs, loudspeakers, ...).
- Absorption per octave band (63 Hz to 8 kHz) and scattering per material, with a library of predefined materials.
- Sources with a level (sound pressure level at 1 m) and several excitation signals; any number of receivers, single or as grids and shells.
- Broadband impulse responses, spectra, the energy decay, and the ISO 3382 parameters T20, T30, EDT, C50, C80, D50 per octave or third-octave band, with export and comparison between simulations.
- Binaural auralization with HRTFs (SOFA files) and live ambisonic playback with head rotation, including a first-person *receiver view*.
- Animations of the simulated pressure field on slice planes, a didactic particle view, and a modes view on sheets at listener height above every floor level.
- Ray-path tracking for checking reflections from individual surfaces.

How this manual is organized

Chapter 2 explains how to install, update and remove F.A.S.E.S. Chapter 3 walks through a first simulation from start to finish. Chapter 4 summarizes the methods, so that you can choose the settings with confidence. Chapters 5 to 12 describe the program window by window. Chapter 13 documents a project folder and all files. Chapter 14 collects tips and solutions to common problems, and the appendices give command-line usage and the licence.

Chapter 2

Installation

2.1 System requirements

Operating system	Linux, 64-bit (x86-64), glibc 2.31 or newer. Tested on Debian 11; any current distribution works (Debian 11 or newer, Ubuntu 22.04 or newer, Linux Mint 21 or newer, Fedora, openSUSE, Arch). On Windows, F.A.S.E.S can run in WSL 2 (Section 2.6; not tested).
Graphics card	An NVIDIA GPU with driver version 450.80 or newer. The CUDA toolkit is *not* needed: the solver contains the CUDA runtime. Maxwell (GTX 900) and newer GPUs are supported.
Python	Python 3.9 or newer (installed automatically by the installer where the distribution provides it).
Memory	8 GB RAM or more recommended. GPU memory limits the largest wave-based model; the program tells you when a simulation does not fit.
Disk	About 1.5 GB for the program and its Python packages, plus space for the simulation results (from a few MB to several GB per project, depending on the model and the animations).

Note. Ubuntu 20.04 has a recent enough glibc but only Python 3.8; the installer stops there with a clear message. Use Ubuntu 22.04 or newer.

2.2 Installing with `install.sh`

F.A.S.E.S is installed with one script, `install.sh`, which you download from the project's *Releases* page:

https://github.com/vitorotri/fases/releases

1. Download `install.sh` from the latest release.
2. Make it executable. In a terminal:

   ```
   chmod +x install.sh
   ```

   or in the file manager: right-click, **Properties**, **Permissions**, tick **Allow executing file as program**.

3. Run it: double-click it (choose **Run** or **Run in Terminal** if asked), or in a terminal:

```
./install.sh
```

4. When asked, type your password (the system packages are installed with `sudo`).
5. Wait for `Done`. The Python packages take a few minutes the first time. The window stays open at the end until you press Enter.

Note. GNOME's file manager (Nautilus) opens scripts in a text editor instead of running them. There, right-click the file and choose **Run as a Program**, or use the terminal.

What the installer does

1. Checks the computer: 64-bit Linux, glibc version, and the NVIDIA driver. A missing or too old driver only produces a warning; the program is installed anyway, but simulations need the driver.
2. Installs the required system packages with the distribution's package manager (apt, dnf, pacman or zypper): Python with `venv` and `pip`, the PortAudio sound library, the OpenGL libraries, and the X11 libraries Qt needs.
3. Downloads the latest release: the program files and the prebuilt solver.
4. Copies everything into `~/.local/share/fases`.
5. Creates a private Python environment there and installs the Python packages listed in `requirements.txt`. Nothing is installed into the system's Python.
6. Creates the command `fases`, an entry in the applications menu, a desktop icon, and an uninstaller.

Important. The installer never installs graphics drivers. If it warns about the NVIDIA driver, install the driver with your distribution's driver manager (for example **Driver Manager** in Linux Mint, **Additional Drivers** in Ubuntu), reboot, and run `nvidia-smi` to check it.

2.3 Starting F.A.S.E.S

- From the applications menu: **F.A.S.E.S.** (category Science or Education; typing `fases` in the menu search finds it).
- From the desktop icon.
- From a terminal: `fases`. If the terminal does not find the command, log out and in again once, so that `~/.local/bin` is on your `PATH`.

2.4 Updating

Run `install.sh` again. It downloads the newest release and replaces the program files. The Python environment is kept, so an update is quick. Your projects are never touched: they live in your own folders, not in the installation.

2.5 Uninstalling

```
~/.local/share/fases/uninstall.sh
```

This removes the program, its Python environment, the `fases` command, the menu entry, the desktop icon and the icons. Your projects are not touched. System packages installed during the installation (Python, PortAudio, ...) stay, since other programs may use them.

2.6 Windows: running F.A.S.E.S in WSL

On Windows, F.A.S.E.S runs in the *Windows Subsystem for Linux* (WSL 2): a Linux system inside Windows, with access to the NVIDIA GPU (CUDA) and to the Windows desktop for its window and sound (WSLg). The Linux version is installed there with the same `install.sh`.

Important. Running F.A.S.E.S in WSL has not been tested yet. The steps below follow the documented behaviour of WSL 2 and of the Linux installer, but this release has only been tested on native Linux. Please report your experience, whether it works or not.

Requirements

- Windows 11 (recommended), or Windows 10 version 22H2 (build 19044 or higher, with the Windows updates from November 2022 or later). Check the version with `Win`+`R`, `winver`.
- An up-to-date NVIDIA driver *for Windows*. It includes the CUDA support for WSL. Do *not* install an NVIDIA driver inside WSL.
- WSL 2 with WSLg and Ubuntu 22.04 or 24.04.

Setting up WSL

In Windows PowerShell:

```
wsl --install -d Ubuntu-24.04     # first time: installs WSL and Ubuntu (restart
    when asked)
wsl --update                      # existing WSL: update it
wsl --version                     # must list a "WSLg version": the graphics support
wsl -l -v                         # your Ubuntu must show VERSION 2
```

At the first start, Ubuntu asks for a user name and a password for the Linux system.

Checking the GPU and the graphics

In the Ubuntu terminal:

```
nvidia-smi                            # must list your GPU (CUDA works)
sudo apt update && sudo apt install -y x11-apps mesa-utils
xeyes                                 # a small window must open (graphics work)
glxinfo -B | grep -i renderer         # "D3D12 (NVIDIA ...)" = GPU-accelerated 3D view
```

If the renderer reads `llvmpipe`, the 3D view works but slowly (software rendering); update the Windows graphics driver.

Installing

In the Ubuntu terminal:

```
curl -LO https://github.com/vitorotri/fases/releases/latest/download/install.sh
chmod +x install.sh
./install.sh
```

The installer works exactly as on Linux (Section 2.2). Start F.A.S.E.S with the command `fases` in the Ubuntu terminal; WSL usually also adds **F.A.S.E.S.** to the Windows Start menu, in the folder of the Ubuntu distribution. There is no desktop icon (WSL has no Linux desktop). Updating and uninstalling work as on Linux.

Working in WSL

- **Keep your projects in the Linux file system**, for example in `~/projects`. Folders on the Windows drives (`/mnt/c/...`) are reached much more slowly, and simulations write many files.
- The Linux files are visible in Windows Explorer under `\\wsl$\Ubuntu-24.04\home\<user>`, to copy models in and results out.
- Sound (auralization) should play through Windows' audio output automatically.
- The GPU is shared with Windows: close GPU-heavy Windows programs before large simulations.

Problems in WSL

`nvidia-smi` fails or lists no GPU
Update the Windows NVIDIA driver, run `wsl -update` and `wsl -shutdown` in PowerShell, and open Ubuntu again.

No window opens
WSLg is missing: check `wsl -version` (it must list a WSLg version). On Windows 10, update Windows to 22H2 first, then run `wsl -update`.

The 3D view is slow
Check the renderer as above; update the Windows graphics driver.

2.7 Running from the source code (developers)

The repository has this layout:

```
fases/
  install.sh  requirements.txt  LICENSE.md
  common/            gui.py and the Python tools (allIRs.py, acouparams.py,
    binauralize.py, ambisonic.py, modes3d.py, fminconFIT.py, sketch2d.py,
    props3d.py, ...), materialsREF.txt, icons/
  3D-solver/
    core/            main3D.cu, functions3D.cuh
    include/         parsing3D.h, ism3D.h, raytrace3D.h
    Makefile
    build/fases3D    the compiled solver
```

Python environment

```
python3 -m venv ~/fases-venv
~/fases-venv/bin/python -m pip install --no-compile -r requirements.txt
~/fases-venv/bin/python common/gui.py
```

The option `-no-compile` avoids a known crash of Python 3.9 while installing PySide6.

Compiling the solver

The solver needs the CUDA toolkit (nvcc 11.2 or newer) and a C++17 compiler.

```
cd 3D-solver
make            # quick build for the GPU set in the Makefile (ARCH)
make release    # portable build for all GPU generations (several times slower to
    compile)
make clean      # remove the binary
```

`make release` builds native code for the GPU generations Maxwell to Ampere, plus generic code that newer GPUs (Ada, Hopper, Blackwell) compile once at the first start. Use it for binaries you give to other people. Run `make` with the same user that can run `nvcc` on your machine (for example with `sudo` if CUDA is only on root's `PATH`).

The program finds the solver automatically: `3D-solver/build/fases3D` next to the `common/` folder, else a `fases3D` next to `gui.py`, else a `fases3D` on the `PATH`.

Chapter 3

Quick start

This chapter runs a complete first simulation. Each step is explained in detail in the later chapters.

1. **Create a project.** In the **Scene** tab, click **New** and choose an empty folder. F.A.S.E.S creates the project structure in it (Section 13.1).
2. **Import the scene.** Click **Import OBJ** and choose your model. Each layer (group) of the OBJ becomes one material row in the table. The model must be in metres.
3. **Set the materials.** For each layer, type the absorption coefficients per octave band and the scattering coefficient, or right-click the row and choose a predefined material.
4. **Place sources and receivers.** In the **Sources** tab, type the coordinates and the level $L_{p,1\,\mathrm{m}}$ of each source, or press S over the 3D view and click to place them. Do the same for receivers in the **Receivers** tab (R).
5. **Set up the simulation.** In the **Setup** tab, check the temperature, then click **Estimate**. It measures the space and fills in suggested values for the grid spacing, the cutoff frequency, the simulated duration and the number of rays. Adjust if needed.
6. **Fit the impedances and simulate.** In the **Run** tab, click **Fit impedances** (converts the absorption of each material into the solver's boundary model), then **Simulate**. The progress and the solver's messages appear in the Output log.
7. **Look at the results.** Switch to **Results** with the mode selector at the top right and click receivers in the scene. Choose what to show at the top of the Results window: the impulse response and spectrum, the energy decay, an acoustic parameter, or the summary.
8. **Listen.** In the **Receivers** tab, open **Auralize**, choose an HRTF file (SOFA) and an anechoic recording, and click **Convolve**, then play. Or tick **Ambisonic live** and click **Live** to turn your head with the dials while listening at a selected receiver.

Chapter 4

How F.A.S.E.S simulates

You do not need these details to use the program, but they explain what the settings do and help you judge the results.

4.1 The hybrid scheme and the crossover frequency

Below the crossover frequency, the impulse response comes from the wave-based solver (FVM); above it, from geometrical acoustics (GA). The crossover frequency is set by one value, the **Cutoff frequency** in the **Setup** tab:

- The cutoff frequency is *the highest frequency you want simulated wave-based.*
- The crossover frequency is placed at the *upper edge of the octave band* containing it: a cutoff of 251 Hz lies in the 250 Hz octave (177 to 354 Hz), so the crossover is at 354 Hz, the FVM covers everything up to 354 Hz and the GA everything above.
- The FVM grid is made fine enough for twice the crossover frequency, so the wave-based part is accurate right up to the crossover.
- A cutoff below 44.5 Hz (the lower edge of the 63 Hz octave) means *GA only*: no FVM is run at all.

Note. Typing an octave edge directly (177, 354, 707, 1414 Hz) gives the most grid points per wavelength for the computing effort, because the cutoff and the crossover frequency then coincide.

4.2 The wave-based solver (FVM)

The Finite Volume Method (FVM) [1] solves the linear acoustic equations [2] in the time domain on a uniform Cartesian grid, on the GPU, resulting in values of sound pressure p and particle velocity $\mathbf{u}$ at the center of each cell of the grid, i.e.,

$$\dot{p} = -\rho_0 c^2 \nabla \cdot \mathbf{u}$$
$$\dot{\mathbf{u}} = -\frac{1}{\rho_0} \nabla p$$

for a medium at rest density ρ_0 and speed of sound c. Surfaces are represented as immersed boundaries on that grid (they do not need to follow the grid lines, drastically reducing staircasing artifacts of rectilinear grids), and each material's frequency-dependent impedance is modelled with a small set of complex poles (using *auxiliary differential equations*, ADE [3]) fitted to the material's absorption (**Fit impedances**). The outer boundary of the computational box can be made absorbing with perfectly matched layers (PML), face by face, for open or outdoor scenes.

Grid resolution

The grid spacing h determines the accuracy. A common rule is at least 10 grid points per wavelength at the highest wave-based frequency, f_{max}:

$$h \leq \frac{c}{10\, f_{\max}}.$$

Finer grids cost memory and time: halving h multiplies the number of cells by 8 and the computing time is raised.

Source signals

- **PCS** (default): the Physically Constrained Source (PCS) [4] is a pulsating-sphere source producing a smoothed flat broadband pulse without DC components and resonant frequency set by **Frequency** (recommended to use 25 Hz).
- **Ricker**: a Ricker wavelet with peak frequency set by **Frequency**.
- **Sinusoid**: a continuous tone per source (column f_0), for studying single frequencies.

4.3 Geometrical acoustics (GA)

Image sources

The image-source method (ISM) [5] finds every specular reflection path up to the **ISM max order** exactly, with angle-dependent reflection coefficients from the material absorption. If the image tree for the requested order would not fit in half the free memory, the order is lowered and the log says so (lines starting with `[!]`).

Ray tracing and scattering

Rays leave each source in all directions and bounce through the scene until they leave the simulated time span or escape through an open side. They carry the specular reflections of higher order than the ISM handles above the crossover time, and all scattered energy. The two parts never overlap: a specular path belongs to the ISM if its order is at most the ISM order, and to the ray tracer otherwise.

Scattering uses two techniques, each doing a different job:

- **On/off scattering moves the energy.** At each reflection, a ray continues in a random (Lambert) direction with probability s (the material's scattering coefficient), otherwise specularly. It always keeps its full energy, so multiple scattering and the decay come out right [6].
- **Diffuse rain estimates the detection.** At each reflection, the expected energy of a scattered ray is deposited directly at every visible receiver [7].

They are matched in one place: the receiver sphere ignores the segment right after a diffuse reflection, because the rain already accounted for it. The totals are the same as with on/off scattering alone, but with much less noise, so the early scattered energy is smooth even with few rays.

Air absorption

Air absorption follows ISO 9613-1 for frequencies above the *cutoff* only (for now), from the temperature and the altitude, with a default relative humidity.

4.4 Combining the parts

The FVM result is resampled from the solver's time step to 44.1 kHz, the sample rate of the GA, and filtered below the crossover frequency; each GA octave band is filtered within its own band. The parts are then added into one broadband impulse response. Both parts are absolute and pre-calibrated on the methods: the impulse responses are in pascals, for a source with the level $L_{p,1\,\mathrm{m}}$ you set, and spectra are in dB with reference at 20 µPa. In free field, the direct sound at 1 m reads exactly $L_{p,1\,\mathrm{m}}$.

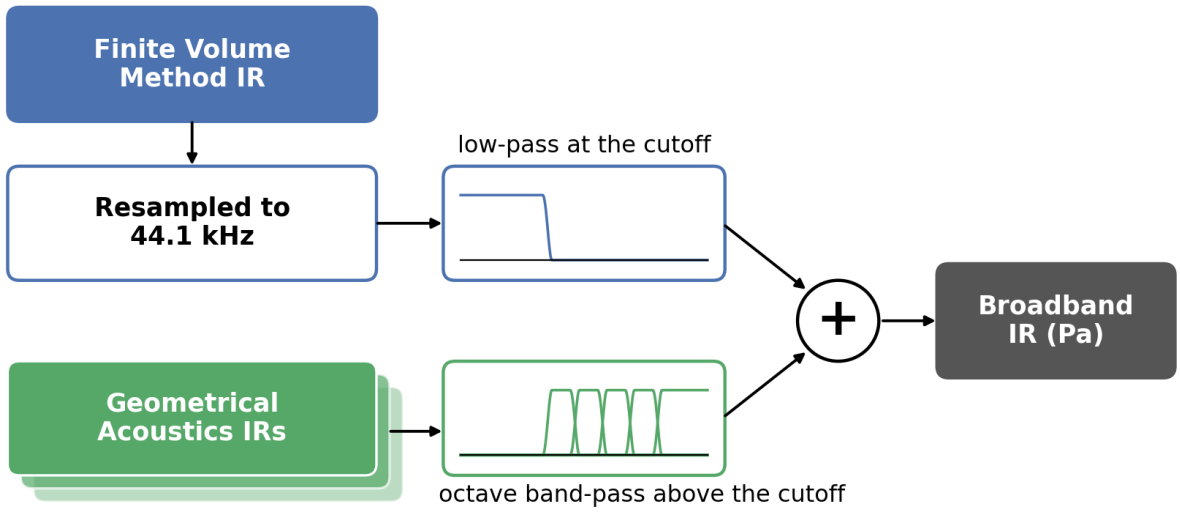


Figure 4.1: Dataflow for the resulting broadband impulse response (IR) in Pascal (Pa)

4.5 Units and coordinates

All lengths are in metres, times in seconds, pressures in pascal, levels in dB SPL. Coordinates are those of your scene. If the scene extends into negative coordinates, the program shifts it into the positive octant for the solver; the right-click menu of the 3D view lets you display and choose coordinates in either frame (**Scene coordinates**).

Chapter 5

The main window

The window has three parts:

- the left panel with the tabs **Scene**, **Sources**, **Receivers**, **Setup** and **Run**;
- the 3D view of the scene, with sources, receivers, slice planes, overlays and the choice between Dark/Light modes;
- the mode selector at the top right of the 3D view: **Simulation** (editing and running), **Results** (plots) and **Draw** (sketcher and props).

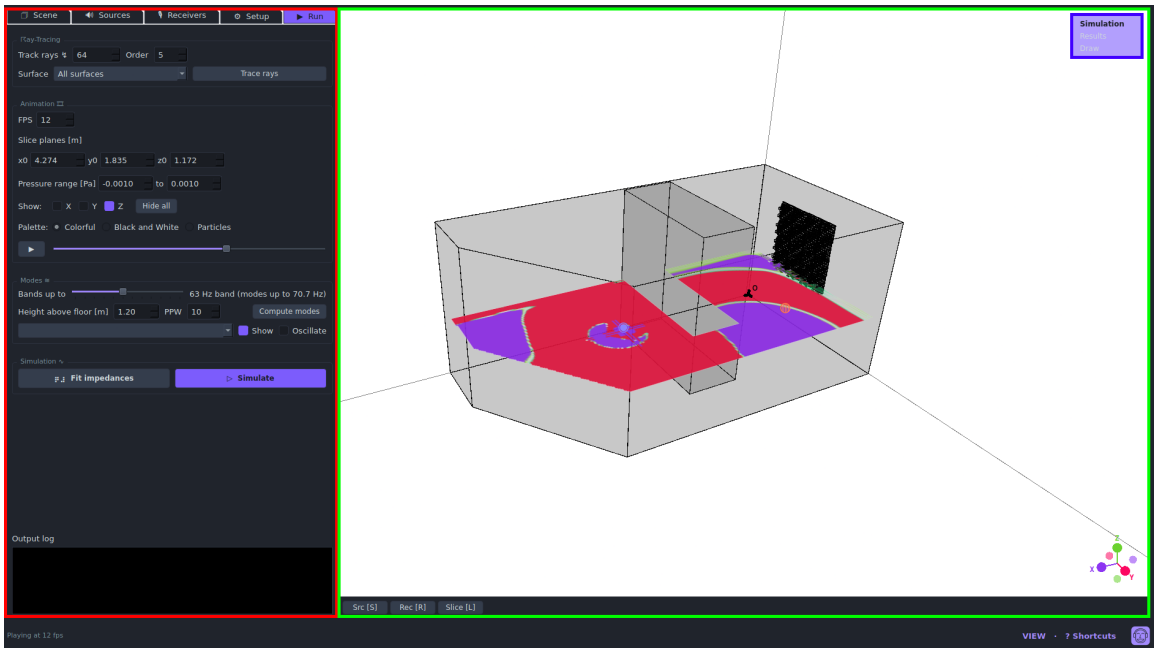


Figure 5.1: F.A.S.E.S main window

The status bar at the bottom shows messages, the current interaction mode, a reminder of the shortcut overlay, and the **About** button (the F.A.S.E.S logo).

5.1 Navigating the 3D view

Input	Action
Left mouse drag	Orbit around the scene
Middle mouse drag	Pan

Input	Action
Mouse wheel	Zoom
Ctrl +left drag	Roll
Alt +left drag	Free tumble
Shift +left drag	Look around without limits
T	Reset the view
Space	Navigation mode: fly with W/X (forward/back), A/D (sideways), Q/Z (up/down), the mouse looks around
Esc	Leave the current mode, or cancel
Tab, Shift+Tab	Cycle through the panel tabs
Right click	Scene menu (below)
?	Show or hide the shortcut overlay

5.2 The scene menu (right click on 3D view)

- **Select surfaces**: click surfaces to select them, or drag boxes around several; then right-click again to **Group** them into one layer, **Explode** a layer into its pieces, or **Cleanse phantom layers** (remove layers without faces).
- **Scene coordinates**: show and enter coordinates in the original OBJ frame or in the frame shifted to the positive octant.
- **Show / Hide axes**, **Light / Dark scene**.
- **Dollhouse view**: removes the walls facing you and the objects in the front half of the scene, so you look into it from outside. +/− (or PgUp/PgDn) raise and lower a horizontal section plane that cuts the scene from the top (100 % = no cut).
- **Top-down (ortho) view** / **Perspective view**.
- **Navigation mode (WAXD fly)**.
- **Enter receiver view**: puts you inside the selected receiver, looking in the direction of its head pose (Section 11.4).

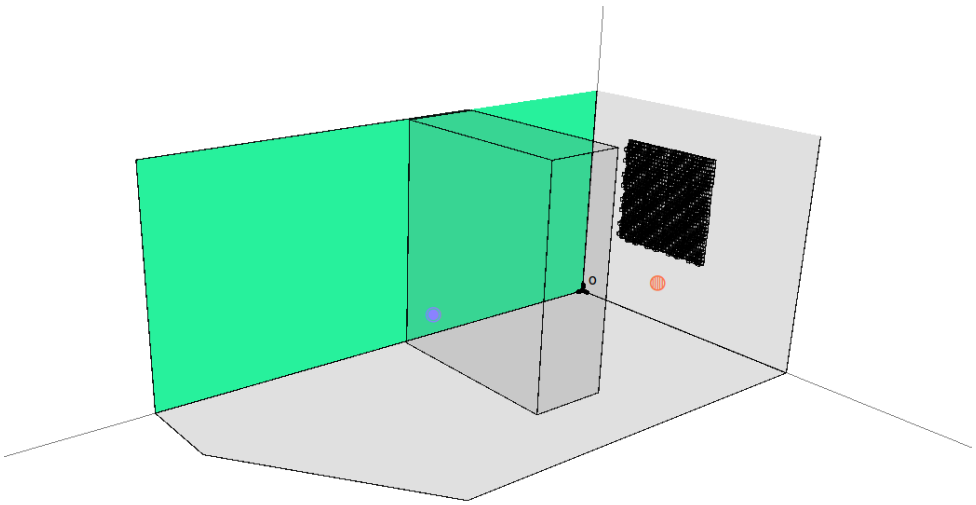


Figure 5.2: Dollhouse view with a selected surface.

Chapter 6

Scene and materials

6.1 Projects

The **Project** box at the top of the **Scene** tab manages the working project folders:

- **New**: choose an empty folder; the project structure is created in it.
- **Open**: open an existing project folder. Everything is restored: scene, materials, sources, receivers, setup, inactive layers and the last results.
- **Save**, **Save as...**: save the project (**Save as...** copies it to a new folder).

A project is a self-contained folder; see Chapter 13 for its contents.

6.2 Importing the scene

Import OBJ loads a Wavefront OBJ model. Requirements:

- **Units in metres.**
- **One layer per material.** Each group (`g` or `o` name) becomes one row of the material table. Several separate objects may share a layer.
- **Closed by surfaces or by PML faces.** The air volume must be enclosed, either by surfaces (floor, walls, ceiling) or by faces of the bounding box with the PML on (Section 8.2). An indoor room is closed by its own surfaces; a stage, a courtyard or a square needs only its ground and the surrounding surfaces, with PML faces on the open sides and the top.
- **Bounding box.** A layer whose name contains `BoundingBox` defines the computational box; it is not a material and is hidden by default (**Show bounding box**).

Note. Models from SketchUp, Rhino, Revit or Blender can be exported to OBJ. Assign one material or layer per acoustic surface type before exporting; it saves work in the material table.

6.3 The material table

Each row is one layer. The columns are: the layer name, the absorption coefficients α for the octave bands 63 Hz to 8 kHz, and the single-number scattering coefficient (**Scatter**).

- **Editing values**: click a cell and type. Only numbers are accepted; absorption and scattering cannot be negative. The associated surfaces are highlighted on the scene.
- **Absorption curve**: selecting a row draws its absorption above the table; drag the points of the curve to change the values.

- **Predefined materials**: right-click a row, **Choose material from predefined list…**. The list comes from `common/materialsREF.txt` (one material per line: name and eight absorption values), which you can extend.
- **Fill values**: right-click a cell, **Apply value to rows above / below** or **to columns behind / after**.
- **Paste**: Ctrl+V pastes tab- or space-separated values, for example a block copied from a spreadsheet, starting at the selected cell.
- **Active layers**: the check box in the name cell switches a layer off. An inactive layer is hidden in the scene and left out of the simulation (useful for comparing variants, such as with and without a reflector). **Deactivate all** / **Activate all** switches all layers at once.
- **Rename / delete**: double-click a name to rename the layer, or right-click **Rename layer…** / **Delete layer…**. Giving a layer the name of another one merges them into one material layer.
- **Save all .mat**: writes the material files. They are also saved automatically before a simulation.
- **Hard / soft switch**: chooses if reflection for fitted boundary has inverted fase - **Soft ($R < 0$)** implies in fase inversion.

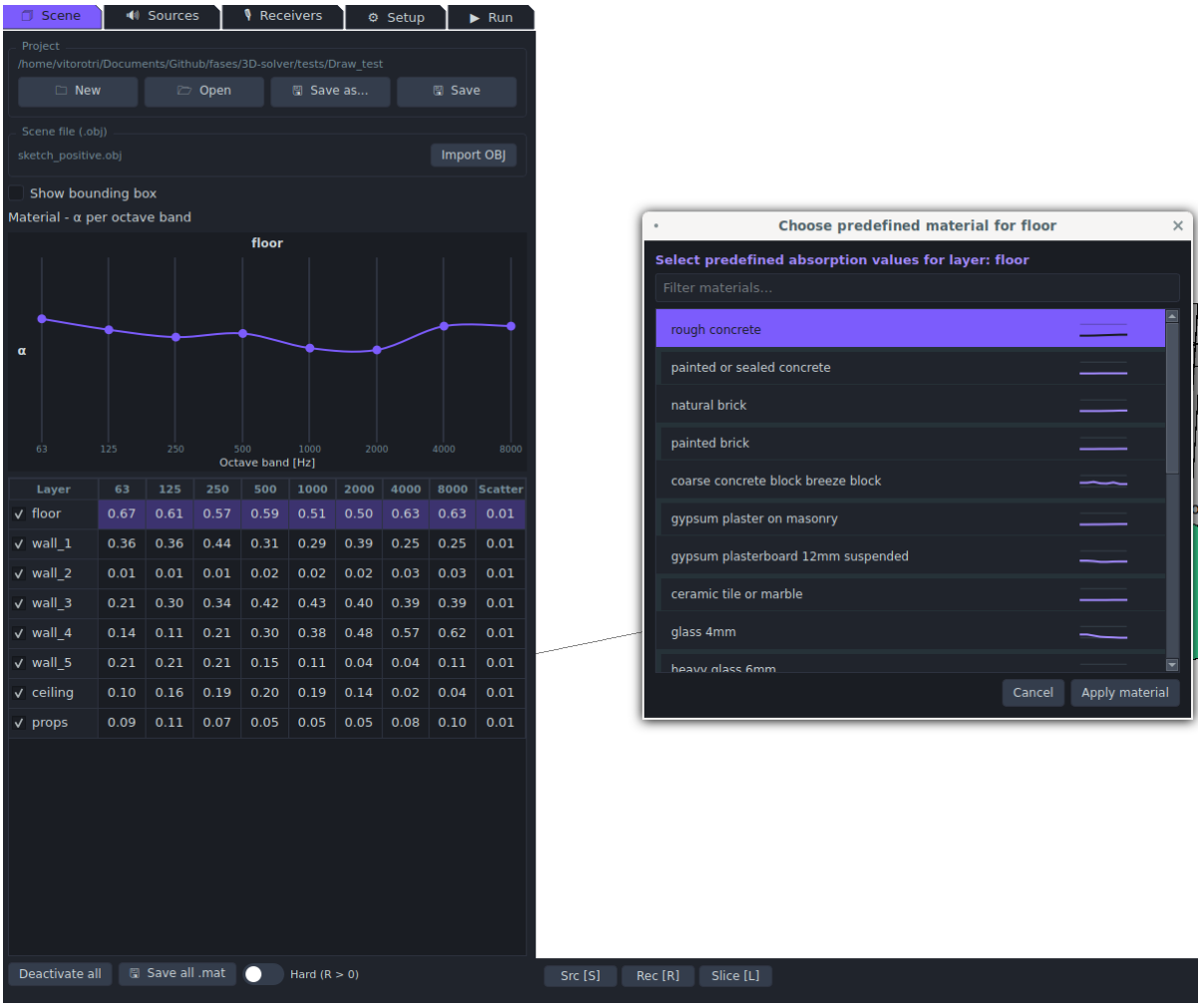


Figure 6.1: Absorption curve editor and material table.

Absorption and scattering

- The absorption coefficient α (0 to 1) is the fraction of incident energy a surface absorbs, per octave band. Use measured random-incidence values where available.
- The scattering coefficient s (0 to 1) is the fraction of reflected energy sent in random directions instead of the mirror direction. Smooth large surfaces: 0.05 or less; seating, coffered ceilings, diffusers: 0.3 to 0.8. One value per material is used.
- An absorption above 0.99 is treated as fully absorbing at every angle of incidence.

Chapter 7

Sources and receivers

7.1 The point tables

The **Sources** and **Receivers** tabs each hold a table: name, x, y, z (metres), and for sources the level $L_{p,1\,\mathrm{m}}$ (dB) and; for sinusoidal sources, the frequency f_0 per source.

- Names are free text (double-click to edit). Number columns accept numbers only.
- The check box in the name cell switches a point off: it stays in the table but is not simulated.
- **Load** / **Save** read and write the position files `src.txt` and `rec.txt`.
- Right-click a cell: **Apply value to rows above / below**.
- Ctrl+V pastes coordinates from a spreadsheet or a text file; extra lines add new points.

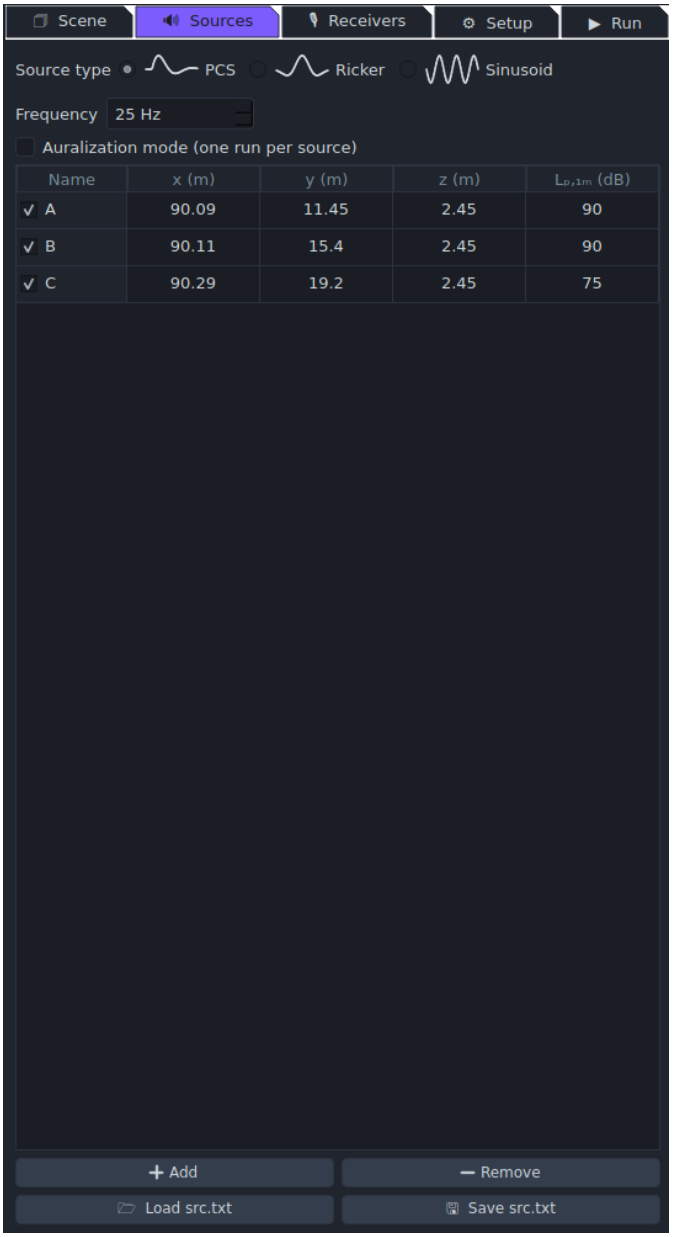


Figure 7.1: The Sources tab.

7.2 Placing points in the 3D view

Input	Action
S / R, then click	Place a source / receiver (also the buttons **Src [S]** and **Rec [R]**)
G while placing	Make a grid or a shell (sphere) array of points
E	Edit mode: drag a placed point, or box-select several
M / R (edit mode)	Move / rotate the selected points
Hold X, Y or Z	Lock movement to that axis
Del, or right-click a point	Delete
Ctrl+Z	Undo

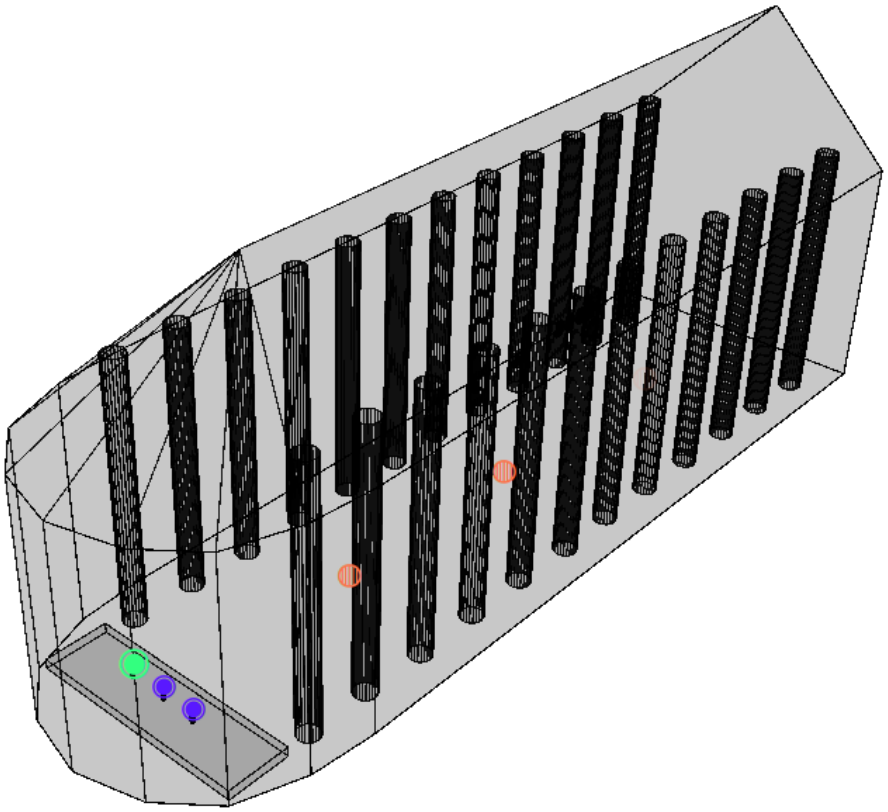


Figure 7.2: Placed sources and receivers on a scene. Sources are shown in blue, receivers in orange, while selected points are highlighted in green.

7.3 Source settings

Below the source table:

- **Source type**: **PCS** (default, for impulse responses), **Ricker** or **Sinusoid**; see Section 4.1.
- **Frequency**: the peak or characteristic frequency of PCS and Ricker Wavelet. It cannot exceed the crossover frequency. For sinusoids, each source has its own frequency in the table instead.
- **Auralization mode (one run per source)**: simulates every source separately instead of all together. Needed when sources should play different signals in the auralization (Section 11.5). The simulation then takes one run per source (longer simulation time).

7.4 Receiver settings

Each receiver also has a head pose (azimuth, elevation, roll) used by the auralization, set with the dials of the **Auralize** panel or in the **Receiver view** (Chapter 11). The **Auralize** button at the bottom of the **Receivers** tab opens that panel.

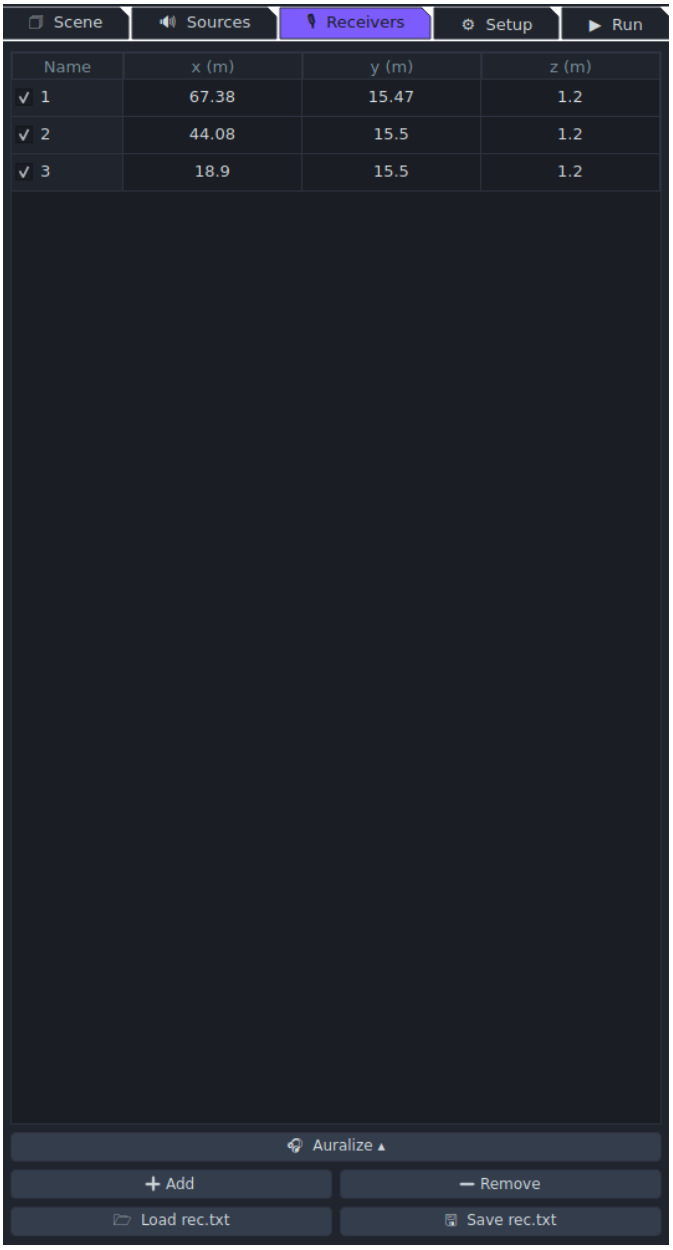


Figure 7.3: The Receivers tab.

Chapter 8

Setup



Figure 8.1: The Setup tab.

8.1 Fields

Field	Meaning
Temperature	Air temperature (°C). Sets the speed of sound and the air absorption.
Altitude	Height above sea level (m). Sets the air pressure for the air absorption.
Estimate	Measures the space (volume, reverberation time estimate, Schroeder frequency) with the solver, without simulating, and fills in suggested values for the fields below. Set the temperature first.
Grid spacing	FVM cell size h (m). Smaller is more accurate but more expensive.
Simulation span	Length of the impulse response (s). Choose at least the expected reverberation time; longer for decay-time evaluation (T30 needs about 35 dB of decay).
Cutoff frequency	Highest wave-based frequency (Hz); sets the crossover frequency (Section 4.1). Below 44.5 Hz: GA only.
CFL number	Time-step safety factor of the FVM (default 0.4). Lower it only if the solver reports instability.
ADE poles	Number of poles of each material's impedance model (default 8).
ISM max order	Highest reflection order of the image-source method (0 to 10). covered by the ray tracer. Typical: 3 to 5, can get very expensive for large scenes.
Number of rays	Rays per source for the reverberant tail. More rays give a smoother tail and take longer. 0 switches the ray tracer off.

8.2 PML faces

The six check boxes (`x+`, `x−`, `y+`, `y−`, `z+`, `z−`) make the corresponding face of the computational box absorbing (with a convolutional perfectly matched layer). Use them where the space is open to the outside: an open-air theatre or stage, a courtyard, a street or a square (open sides and top), an opening in a model. **Deselect all** / **Select all** switches all faces.

8.3 Saving the setup

Save config.txt and **Load config.txt** write and read the setup file directly. The setup is also saved with the project and before every simulation.

8.4 Choosing the values

1. Set the temperature and altitude, then click **Estimate**.
2. Check the suggested cutoff: a common choice is the Schroeder frequency of the space, above which the GA is reliable. Prefer an octave edge (177, 354, 707 Hz). In an open space (stage, street, square) the estimated reverberation time and Schroeder frequency have little meaning: set the cutoff as high as your GPU allows, since diffraction at edges and openings is what the wave-based part captures.

3. Check the grid spacing against your GPU memory; the solver stops with a clear message if the model does not fit, and the remedy is a larger grid spacing or a lower cutoff.
4. Set the simulation span to the reverberation time or a bit more.

Chapter 9

Running simulations

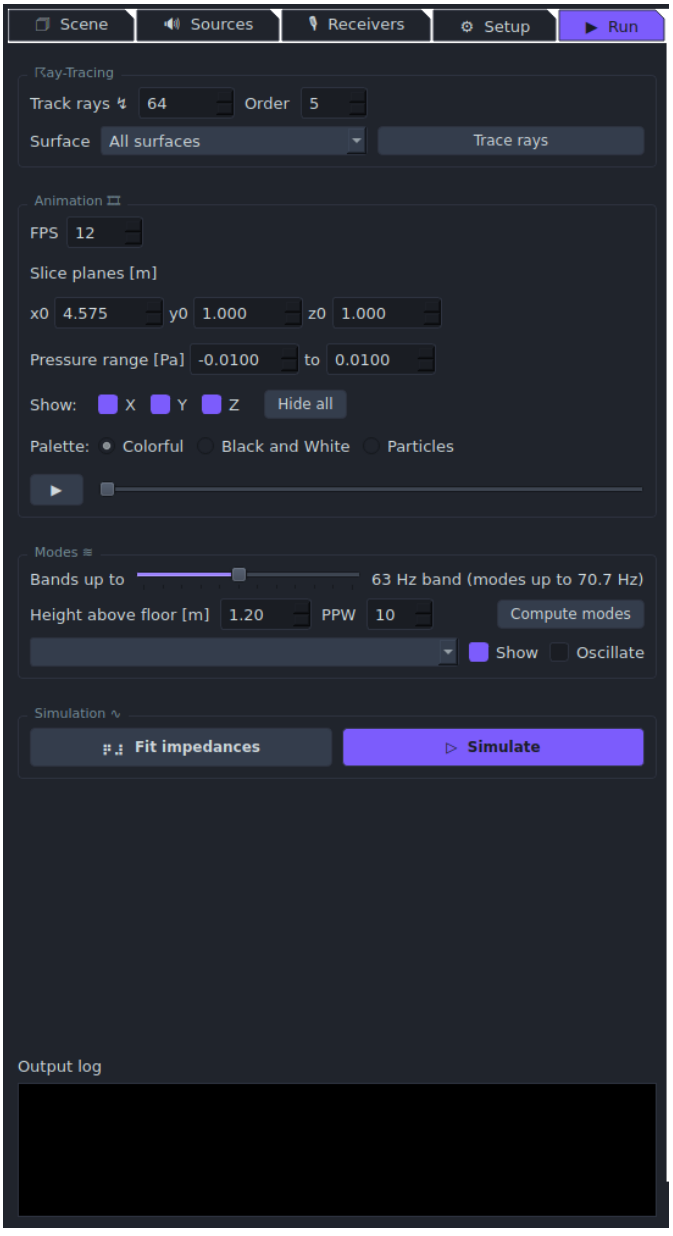


Figure 9.1: The Run tab.

9.1 Simulation

- **Fit impedances**: fits the boundary model (the ADE poles) of every material to its absorption coefficients, up to the crossover frequency. Run it after changing materials, the cutoff, or the number of poles. The results are kept per material and reused while nothing changed.
- **Simulate**: runs the simulation. The button turns into **Stop** while it runs. Progress, time estimates and messages appear in the Output log and in a progress bar over the scene. With **Auralization mode**, the sources are simulated one after another. If relevant values are changed, the button suggests a refit before simulating.

When only geometrical-acoustics settings changed since the last complete run (ISM order, number of rays, scattering), F.A.S.E.S reruns only the GA part and reuses the wave-based results, which is much faster.

9.2 The Output log

The log at the bottom shows every command and the solver's messages. Its height can be dragged at its top edge. Lines starting with `!!` or `[!]` are warnings worth reading, for example when the image-source order was lowered to fit in memory.

9.3 Ray-path overlay

The **Ray-Tracing** box draws ray paths in the scene, for checking reflections:

- **Track rays**: number of rays drawn from each source; **Order**: number of reflections per ray.
- **Trace rays** draws them (coloured by reflection order); click again to clear.
- **Surface**: shows only the rays whose first reflection hits the chosen surface, with that reflection, for example to see where a reflector sends sound.

The overlay is purely visual and does not affect the simulation.

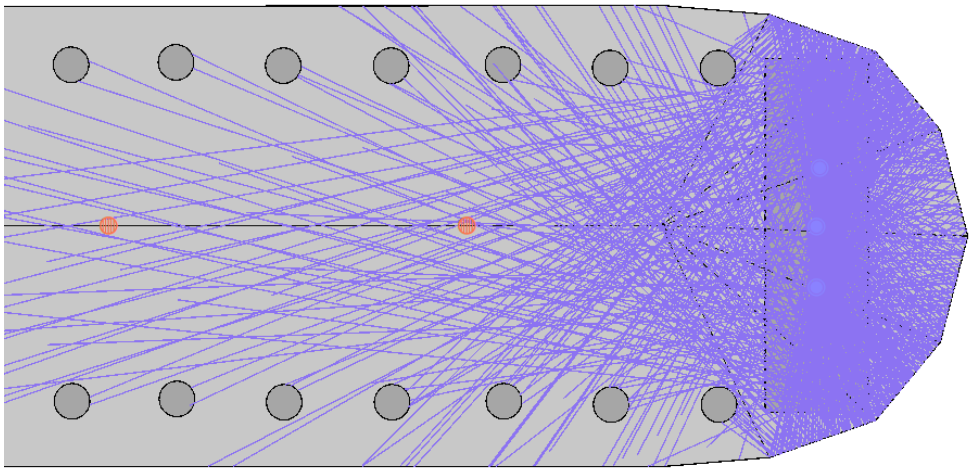


Figure 9.2: Ray paths filtered by surface.

9.4 Animation

After a wave-based simulation, the pressure field can be played back on three slice planes through the scene.

- **FPS**: playback rate (also sets how many frames the solver stores).
- **Slice planes**: positions x_0, y_0, z_0 (m). Or press [L] over the scene, hold [X], [Y] or [Z] and drag to move a plane.
- **Pressure range**: the colour scale (Pa).
- **Show**: which planes are shown; **Hide all** / **Show all**.
- **Palette**: **Colorful**, **Black and White**, or **Particles**.
- Play/pause and the slider: scrub through the frames; the arrow keys step frame by frame.

Particles

The **Particles** palette shows the sound as moving air particles on the slice planes, a teaching view. Particles at rest are faint; they light up and change colour (green-cyan to magenta to amber) where a wavefront passes and where they are displaced. The displacement follows the source level, compressed so that quiet and loud sources both stay visible. This view uses its own fast 2D simulation on each plane and is meant for illustration, not for measurement.

9.5 Modes

The **Modes** box computes the resonances (modes) of the space and shows them on sheets at a chosen height above every floor level, including raked floors, steps, stages and balconies.

- **Bands up to**: a stepped slider over the third-octave bands from 20 Hz to 250 Hz. All modes are computed, from the lowest up to the upper edge of the chosen band.

- **Height above floor**: height of the sheets (default 1.2 m, seated ear height).
- **PPW**: grid points per wavelength (10 to 20). Higher gives finer sheets and more exact frequencies, and takes longer.
- **Compute modes**: runs the calculation in the background (progress in the Output log).
- The list shows each mode with its frequency and band. Picking one shows it: the sheet is lifted where the pressure is positive and lowered where it is negative, coloured red and purple-blue; flat, light areas are the nodes.
- **Show** hides or shows the sheets; **Oscillate** animates the mode slowly.

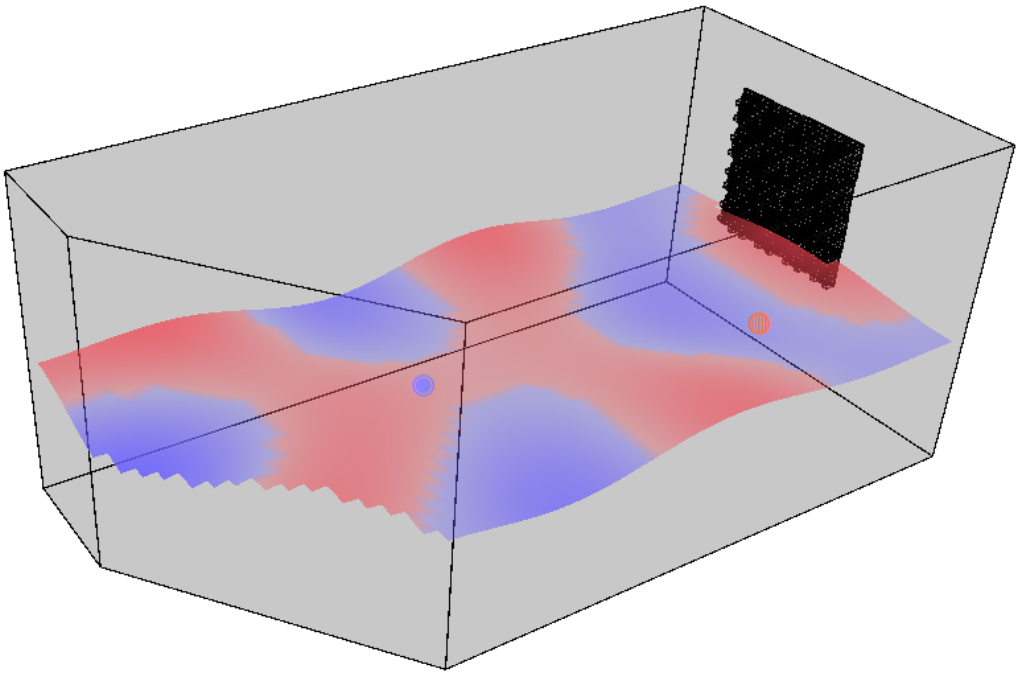


Figure 9.3: A mode on sheets above the floors.

How the modes are computed

The air volume of the scene is divided into a grid of cells (by casting rays through the scene), and the eigenvalue problem of the Helmholtz equation is solved on it. All active layers are included: walls, columns, partitions and balconies act as rigid surfaces, and thin surfaces block sound like walls. Faces of the computational box with the PML on are treated as pressure-release openings ($p = 0$).

Note. The modes view is an ideal, lossless model: materials and absorption are ignored, and an open side is approximated by a pressure-release boundary. It shows where modes form and their frequencies; for the real, damped behaviour use the simulated impulse responses. In a mostly open space (stage, street, square) resonances are weak and the open sides are only approximated, so the view is most telling for enclosed or partly enclosed spaces. The grid is limited to 60 000 cells and 200 modes to keep memory moderate; for large spaces at higher bands the log warns that the grid was coarsened.

Chapter 10

Results

Switch to **Results** with the mode selector. The Results window opens; click receivers in the 3D view to select or deselect them for the plots. After a new simulation, the broadband impulse responses and the parameters are computed automatically the first time you open the results.

10.1 Views

The selector at the top of the Results window chooses what is shown:

- **IR + Spectrum**: the impulse response and its spectrum (dB re 20 µPa) of the selected receivers.
- h^2 **normalized (dB)**: the energy of the impulse response over time, normalized.
- **T20**, **T30**, **EDT**, **C50**, **C80**, **D50**: the ISO 3382 parameter per band for each selected receiver, plus their average (energy average for C50 and C80). The second selector switches between octave and third-octave bands.
- **Summary (avg)**: the single-number values averaged over the receivers (T20, T30 and EDT at 500–1000 Hz; C50, C80 and D50 at 500–2000 Hz).

When a project has several sources, the parameter plots show one panel per source.

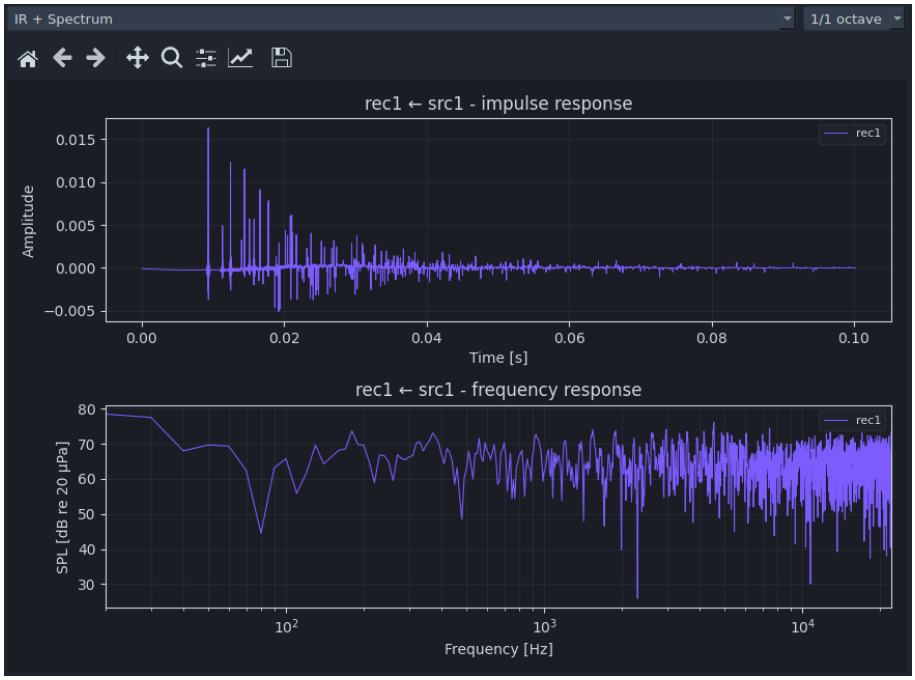


Figure 10.1: Impulse response and spectrum.

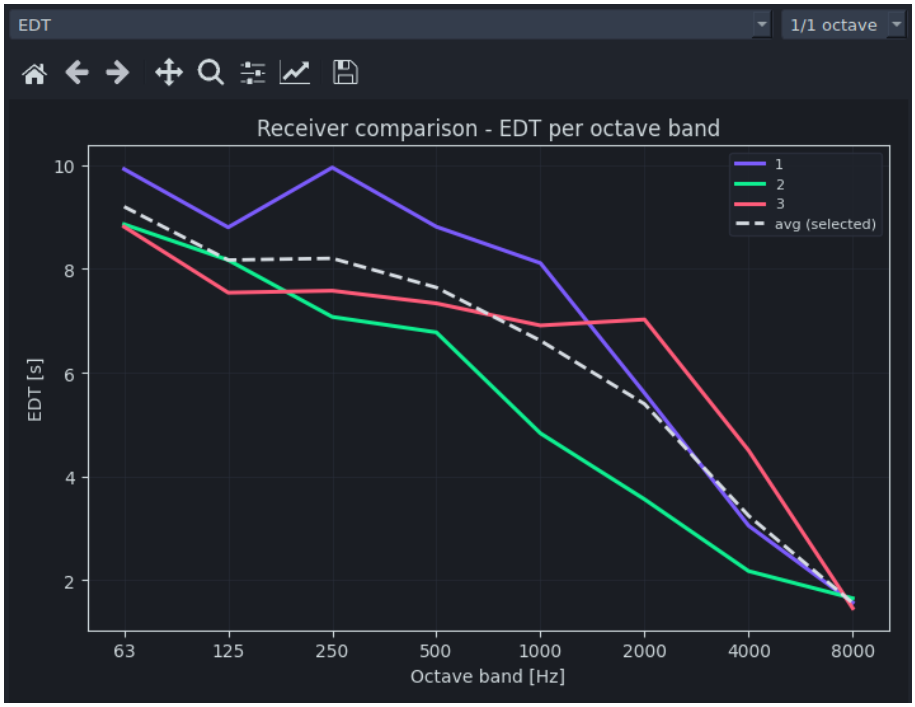


Figure 10.2: An acoustic parameter per band for distinct receivers.

10.2 Zoom, save and export

- The toolbar above each plot zooms (draw a box), pans, goes back and forward, and saves the figure as PNG, SVG or PDF. The cursor position is shown in the toolbar.
- In the parameter and summary views, right-click the plot: **Copy as CSV** (for pasting into a spreadsheet), **Export CSV...**, **Export JSON...**. `Ctrl`+`C` copies too.

10.3 Comparing simulations

In the **Summary** view, **Add simulation...** adds the results of another project folder, so that variants (materials, geometry) can be compared side by side. **Clear compared** removes them.



Figure 10.3: The summary with compared simulations.

Chapter 11

Auralization

Auralization makes the simulated space audible: a dry (anechoic) recording is played through the simulated space, as heard at a receiver. Open the panel with **Auralize** in the **Receivers** tab, and from the table select the receiver to listen at.

11.1 Head pose

The three dials set the listener's head orientation at the selected receiver: **Azimuth** (turning, 0–360°), **Elevation** (looking up or down) and **Roll** (tilting). Each receiver remembers its own pose.

11.2 Binaural rendering (Convolve)

1. **SOFA...**: choose an HRTF data set in SOFA format (head-related transfer functions; free sets are available from several research institutes).
2. **Anechoic...**: choose a dry recording (WAV, FLAC, AIFF or OGG).
3. **Convolve**: computes the binaural impulse response of the selected receiver at its head pose and renders the recording through it. The result is written to `output/binaural_rec<N>.wav`.
4. Play with the play button; the slider seeks. The restore button (circular arrow) returns the dials to the pose the file was rendered with.

Use headphones. The rendered file of every receiver is kept: selecting another receiver loads its file, and if audio was playing it continues at the same point, so you can compare positions.

11.3 Live ambisonic playback

Tick **Ambisonic live** and click **Live**. The sound plays continuously and follows the head pose as you move the dials, without rendering again. With a SOFA file, F.A.S.E.S uses third-order ambisonics decoded with the HRTFs; without one, a simpler first-order rendering. The button pulses red while the audio is prepared and turns green when playing; click it again to stop. The preparation of each receiver is saved (`output/live_cache/`), so starting it again, or switching back to a receiver you already played, starts within about a second. Selecting another receiver while live audio plays restarts the playback for that receiver. The pipeline of the work with the 3rd order Ambisonics is found int the Project's repository in docs/ambisonics_16ch.png.

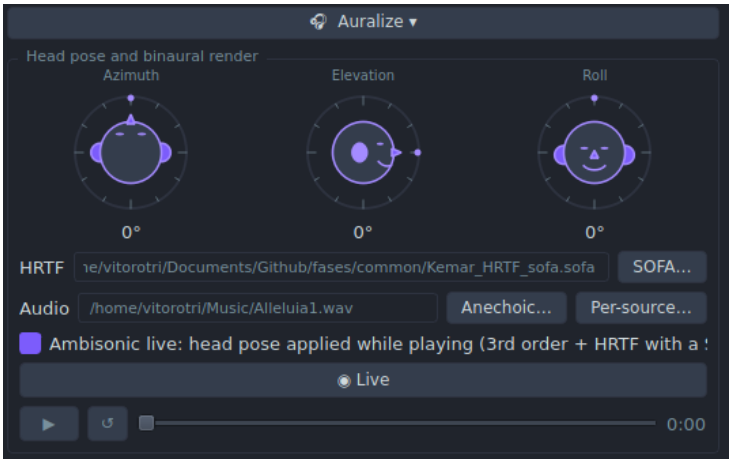


Figure 11.1: The Auralize panel.

11.4 Receiver view

Right-click the scene and choose **Enter receiver view**: the camera moves into the selected receiver and looks in the direction of its head pose. Drag with the left mouse button to look around, `Ctrl`+drag to tilt; the dials follow, and the live audio turns with you. What you see is what you hear. Right-click, **Go back**, to leave.

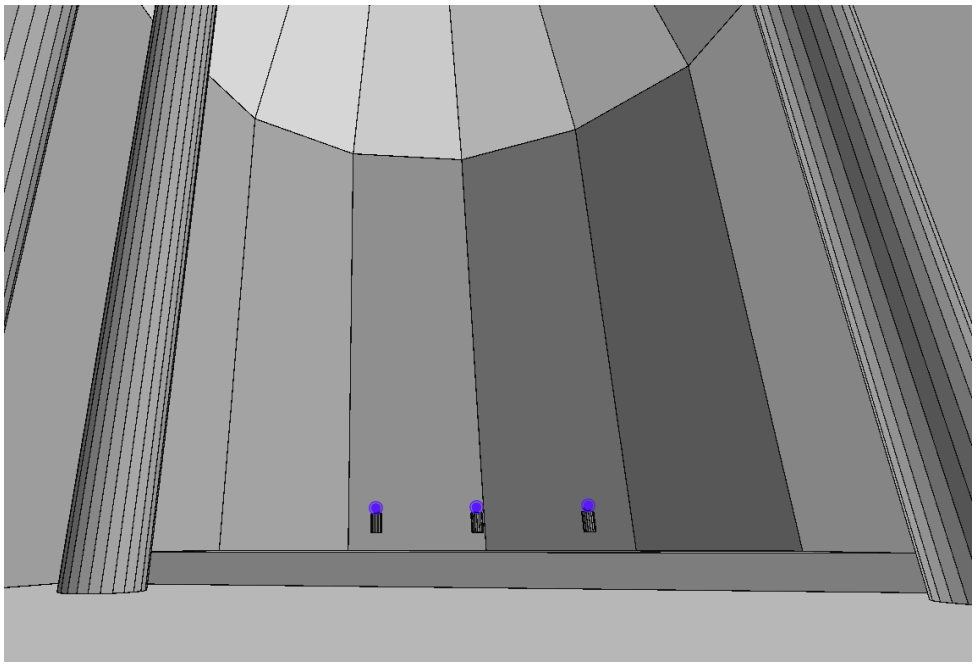


Figure 11.2: Receiver view for a scene inside a gothic cathedral.

11.5 Per-source signals

Per-source... assigns each source its own recording (for example a different instrument or voice per loudspeaker); empty rows use the default recording. This needs the project to be simulated with **Auralization mode (one run per source)** in the **Sources** tab.

The audio settings (recordings, per-source recordings, SOFA file, pose) are saved per receiver as soon as you set them, in `output/auralization.json`, and restored when you select the receiver again.

Chapter 12

Draw mode: sketcher and props

Switch to **Draw** with the mode selector. Two windows open: the **Props** window (top left) and the 2D sketcher (bottom right). The 3D view shows the whole scene from above at first.

12.1 The 2D sketcher

The sketcher draws spaces in plan view and extrudes them into a 3D scene. Every point has a bottom and a top height (z_{base}, z_{top}), so floors and ceilings can slope. A grey top view of the scene loaded in 3D is shown for reference.

Tool	Use
Poly	Click to add points (they snap to the grid); drag a point to move it; Ctrl+K closes the shape; Ctrl+click on an edge inserts a point; Ctrl+right-click deletes a point. Clicking inside a closed shape starts a sub-region; finishing an open line inside a closed shape with Enter shapes the ceiling (raise its z_{top} for a ridge or peak).
Rect, Ellipse	Drag a box or ellipse; overlapping shapes join into one boundary.
Measure	Drag to read a distance.
Select	Box-select points, then Del, or set a height and **Apply z → pts**.
Cut	Drag a line fully across a shape to split it in two.
Shape	Click a shape, then drag to move it, R to rotate, S to scale, Del to remove; name its roof for its ceiling layer. **Convert ⇄ prop** turns a closed shape into a prop and back.
Edge	Drag boxes to select walls, then type a layer name and **Assign → edges**: walls with the same name share one material.

Everywhere: middle mouse drag pans, the wheel zooms, Ctrl+Z undoes. **Import DXF...** loads an outline from a CAD floor plan as the boundary. Right-click: **Extrude to OBJ...** asks for the height and creates the 3D scene; **Update OBJ** rebuilds it after changes.

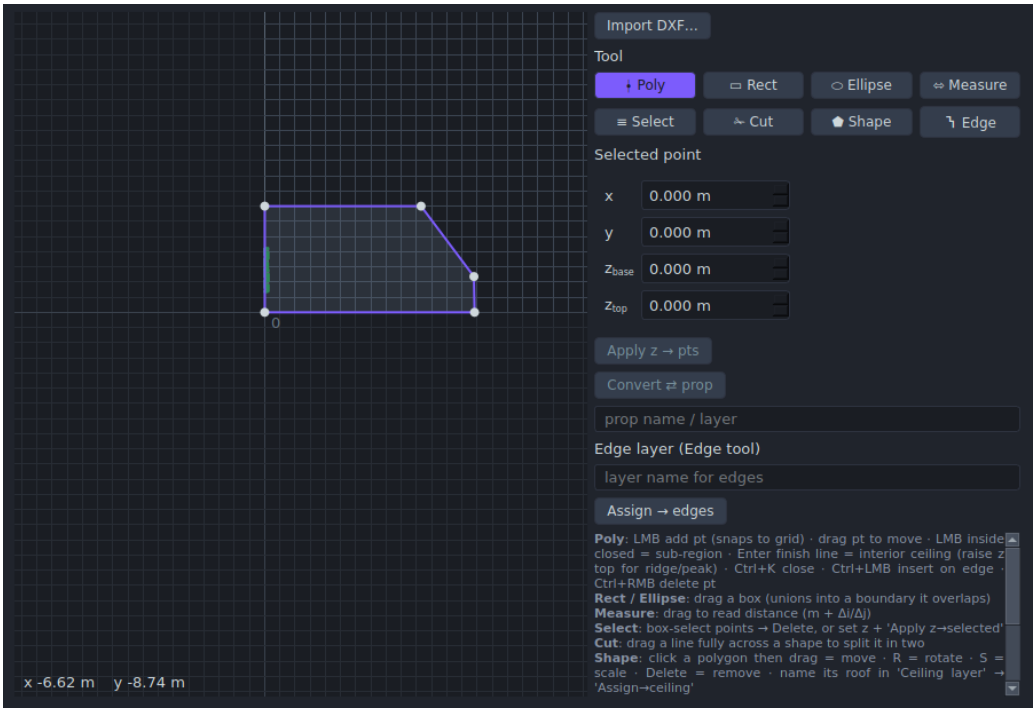


Figure 12.1: The 2D sketcher.

12.2 Props

Props are ready-made 3D objects placed in the scene: `plane`, `box`, `cylinder`, `tetrahedron`, `sphere`, `dodecahedron`, `octant`, `wedge`, `dummy` (a person), `seat`, `sofa`, `bed`, `shelf`, `full_shelf`, `fountain`, `soundbox` (a loudspeaker), `skyline` (a diffuser), `stairs`, and further assets. It is recommended to insert and edit some props in the **Top-down (ortho) view**.

- In the **Props** window, click a primitive under **Add primitive** to place it. Placed props are listed below (`Ctrl`/`Shift`-click to select several); **Delete selected** removes them.
- In the 3D view: `M`, `R`, `S` move, rotate and scale the selected prop; `X`/`Y`/`Z` lock an axis; `Enter` confirms, `Esc` cancels.
- `G` then drag a box: select a group of props, then move, rotate or scale them together.
- `C` then drag a box: copy props.
- Props automatically rest on the surface below them (floor, stage, risers).
- Each prop belongs to a layer; props on the same layer share one material.
- To include props in the simulation, bake them into the scene: right-click in the sketcher, **Update OBJ** (or **Update props in scene OBJ** for an imported scene).

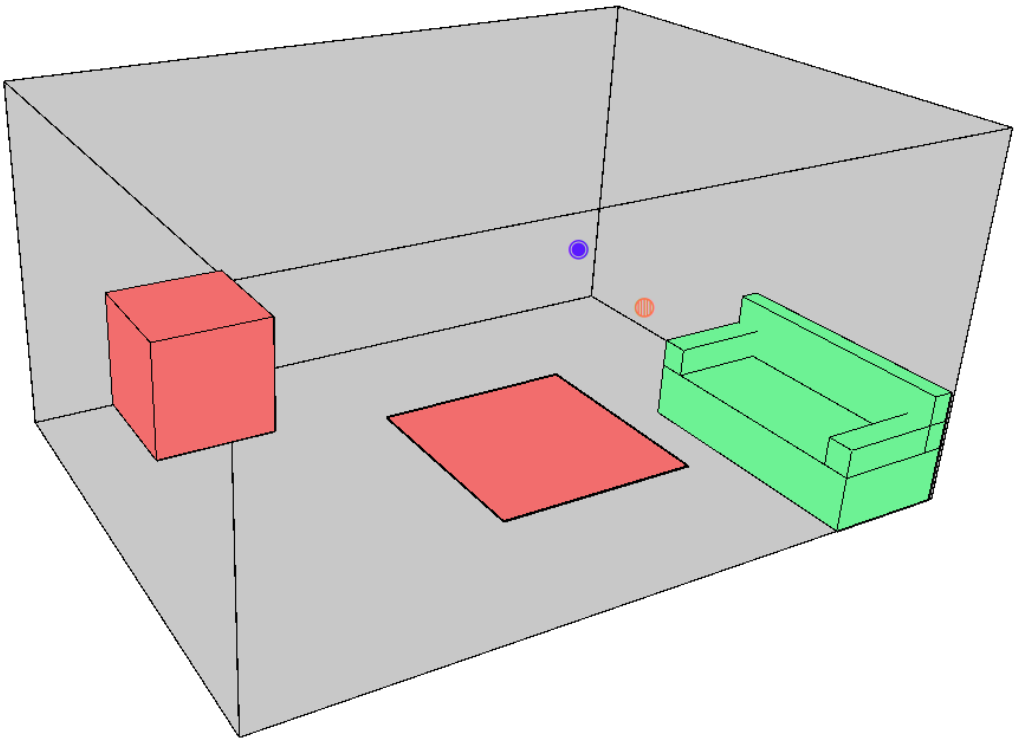


Figure 12.2: Props of different layers (different colors) in a room on **Draw mode**.

Chapter 13

Projects and files

13.1 The project folder

```
my_project/
  project.txt          the project manifest (scene, solver, layers, frames)
  config/
    config.txt         the setup (see the config.txt section below)
    poles.txt          impedance model settings
    p_ref.bin          the source pulse (written by the solver)
  input/
    scene/             the scene OBJ (and derived copies)
    mat/<layer>.mat    absorption per octave band, one file per layer
    fit/<layer>.fit    fitted impedance model per layer
    sctr/<layer>.sctr  scattering coefficient per layer
    src/src.txt        sources (plus .names, .table sidecar files)
    rec/rec.txt        receivers (plus .names, .table, .poses)
  output/              all results (see the output files section below)
```

All files are plain text where possible and can be edited by hand. Lines starting with `#` are comments.

13.2 `config.txt`

One value per line, in this order:

Line	Value
1	Grid spacing h (m)
2	Simulation span (s)
3	Cutoff frequency (Hz), the highest wave-based frequency
4	Temperature (°C)
5	CFL number
6	Number of ADE poles
7–12	PML flags (1 = on) for the faces $z+$, $z-$, $x+$, $x-$, $y+$, $y-$
13	ISM maximum order
14	Altitude (m)
15	Number of rays
16	Source type (0 = PCS, 1 = Ricker, 2 = Sinusoid)
17	Source frequency for PCS / Ricker (Hz)
18	Source index (auralization mode; 0 otherwise)

Line	Value
19–21	Slice plane positions x_0, y_0, z_0 (m)
22	Animation frames per second
23–24	Animation pressure range, minimum and maximum (Pa; display only)

13.3 Sources, receivers and materials

- `src.txt`: one source per line, `x y z Lp [f0]` ($L_{p,1\,\mathrm{m}}$ in dB; f_0 for sinusoids). Only active sources are written; names and inactive rows are kept in the sidecar files.
- `rec.txt`: one receiver per line, `x y z`. Head poses are kept in `rec.txt.poses`.
- `<layer>.mat`: eight absorption coefficients, 63 Hz to 8 kHz.
- `<layer>.sctr`: one scattering coefficient.
- `common/materialsREF.txt`: the predefined material list, one material per line, `name a63 a125 a250 a500 a1000 a2000 a4000 a8000` (underscores in names appear as spaces in the menu).

13.4 Output files

File	Content
`IR<r>_<s>.bin`	Wave-based (FVM) pressure at receiver r for source s
`VEL<r>_<s>.bin`	Particle velocity at the receiver (for the ambisonic rendering)
`HIR<r>_<s>_<band>.bin`	Geometrical-acoustics impulse response per octave band
`HRT<r>_<s>_<band>.bin`	The ray-tracer part of it (auralization tail)
`ARR<r>_<s>.bin`	The image-source arrivals with directions (binaural rendering)
`IR_broad_<r>_<s>.npy`	Broadband impulse response (FVM + GA) in Pa, 44.1 kHz
`params.json`, `params3.json`	Acoustic parameters per receiver, octave / third-octave bands
`header.txt`	Grid and time information of the last wave-based run
`pressure*`, `I*`	Animation slices
`ray_paths.txt`	Ray-path overlay
`binaural_rec<N>.wav`	Binaural renders
`auralization.json`	Auralization settings per receiver
`live_cache/`	Saved live-playback preparations
`modes.npz`	The last modes result

All of `output/` can be deleted; it is recreated by the next simulation. Deleting `live_cache/` only makes the next live start slower.

Chapter 14

Tips and troubleshooting

14.1 Accuracy

- **Grid resolution**: check that the cutoff and grid spacing give at least 10 points per wavelength at twice the crossover frequency; for reverberant spaces use more (Section 4.1).
- **Simulation span**: it must cover the decay you evaluate. T30 needs 35 dB of decay, so the span should be at least the reverberation time.
- **Rays**: if the reverberant tail looks noisy or the parameters vary between receivers close to each other, increase the number of rays.
- **Materials**: the results are only as good as the absorption data. Use measured values; check that every layer has realistic values (an unassigned layer with zero absorption makes the space ring).

14.2 Common problems

The solver stops with a GPU memory error
The wave-based model does not fit in the GPU memory. Increase the grid spacing or lower the cutoff frequency, shorten the simulation span, or close other programs using the GPU.

No NVIDIA driver / the solver does not start
Run `nvidia-smi` in a terminal. If it fails, install or update the NVIDIA driver (version 450.80 or newer) with your distribution's driver manager and reboot.

The installer cannot install a package
Optional X11 libraries that do not exist on your distribution only produce a warning. If a required package fails, install Python 3 (with venv and pip), PortAudio and the GLU library with your package manager and run the installer again.

The program window does not open (Qt platform plugin error)
Install the X11 libraries for Qt, for example on Debian/Ubuntu: `sudo apt install libxcb-cursor0 libxkbcommon-x11-0 libxcb-icccm4 libxcb-keysyms1`.

No sound / "PortAudio library not found"
Install PortAudio: `sudo apt install libportaudio2`. For headphones connected after starting F.A.S.E.S, the player follows the system's default output automatically.

The menu search does not find F.A.S.E.S.
Search for `fases`, or look in the Science or Education category.

Auralization says "no HRT files"
The results come from an older version; run the simulation again (a GA-only rerun is enough).

The parameters show gaps in the lowest band
The impulse response is too short or too noisy at that frequency for the parameter to be defined there; the band is left empty instead of showing a wrong value.

The log reports that the ISM order was reduced
The image tree for the requested order did not fit in memory; the ray tracer covers the remaining orders. Lower the ISM order to avoid the message.

The modes view warns about a coarse grid
The space is large for the chosen band; choose a lower band, or accept approximate frequencies near the top of the band.

14.3 Getting help

Please report problems through the project's issue tracker on GitHub, with the Output log and a description of the steps.

Appendix A

Command-line reference

A.1 The solver

```
fases3D <scene.obj> <config.txt> <src.txt> <rec.txt> [poles.txt] [options]
```

Run from the project folder. Options:

- `-estimate`: measure the space (volume, reverberation time, Schroeder frequency) without simulating.
- `-ga-only`: rerun only the geometrical-acoustics part, reusing the existing wave-based results.
- `-dump-rays N ORDER`: write the ray-path overlay.

A.2 The Python tools

All tools are in `common/` and are normally run by the GUI. Run them with the program's Python (`~/.local/share/fases/venv/bin/python` for an installed copy), from the project's `output/` folder.

- `allIRs.py`: builds the broadband impulse responses.
- `acouparams.py`: computes the acoustic parameters (`-fractions 3` for third-octave bands).
- `binauralize.py`: binaural rendering, `binauralize.py <hrtf.sofa> -rec N -dry <audio> -out <file.wav> -pose AZ EL ROLL`.
- `ambisonic.py`: ambisonic rendering, `ambisonic.py -rec N -dry <audio> [-sofa <hrtf.sofa>] [-live] [-out <file.wav>]`.
- `modes3d.py`: modes; run without arguments for a self-check against the analytical modes of rectangular rooms.
- `fminconFIT.py`: fits the impedance model of the materials.

Most tools run a self-check when started without arguments.

Appendix B

Licence and acknowledgements

B.1 Licence

F.A.S.E.S is distributed under the **PolyForm Perimeter License 1.0.1** (https://polyformproject.org/licenses/perimeter/1.0.1). The full text is in `LICENSE.md` and in the program's **About** window.
Required Notice: Copyright (c) Vito Romanelli Tricanico.

B.2 Third-party software

F.A.S.E.S uses open-source Python packages, which the installer downloads from the Python Package Index under their own licences: PySide6 (Qt), PyOpenGL, NumPy, SciPy, Matplotlib, pyfar, pyrato, scikit-rf, SoundFile, sounddevice, ezdxf, Shapely and mapbox-earcut. The solver contains the NVIDIA CUDA runtime, distributed under NVIDIA's licence terms.

B.3 Citing F.A.S.E.S

If you use F.A.S.E.S in published work, please cite:

> *V. Romanelli Tricanico, F.A.S.E.S: Faithful Acoustic Solver for Extended Spaces, version 0.9, 2026. https://github.com/vitorotri/fases*

Appendix C

Miscelaneous Results

Some of the results on the figures below can be found with their respective data in the project's repository under 3D-solver/tests.

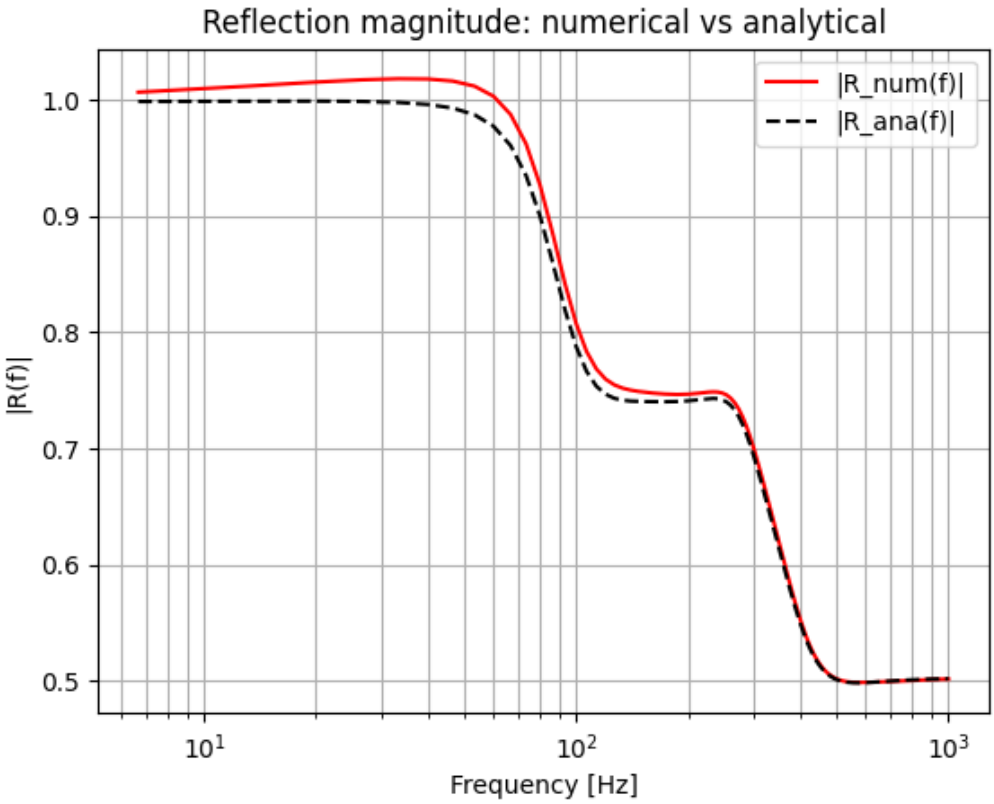


Figure C.1: Comparison of the numerical and expected reflection coefficient measured at a receiver near a wall with impedance matching the analytical reflection, inside an anechoic domain.

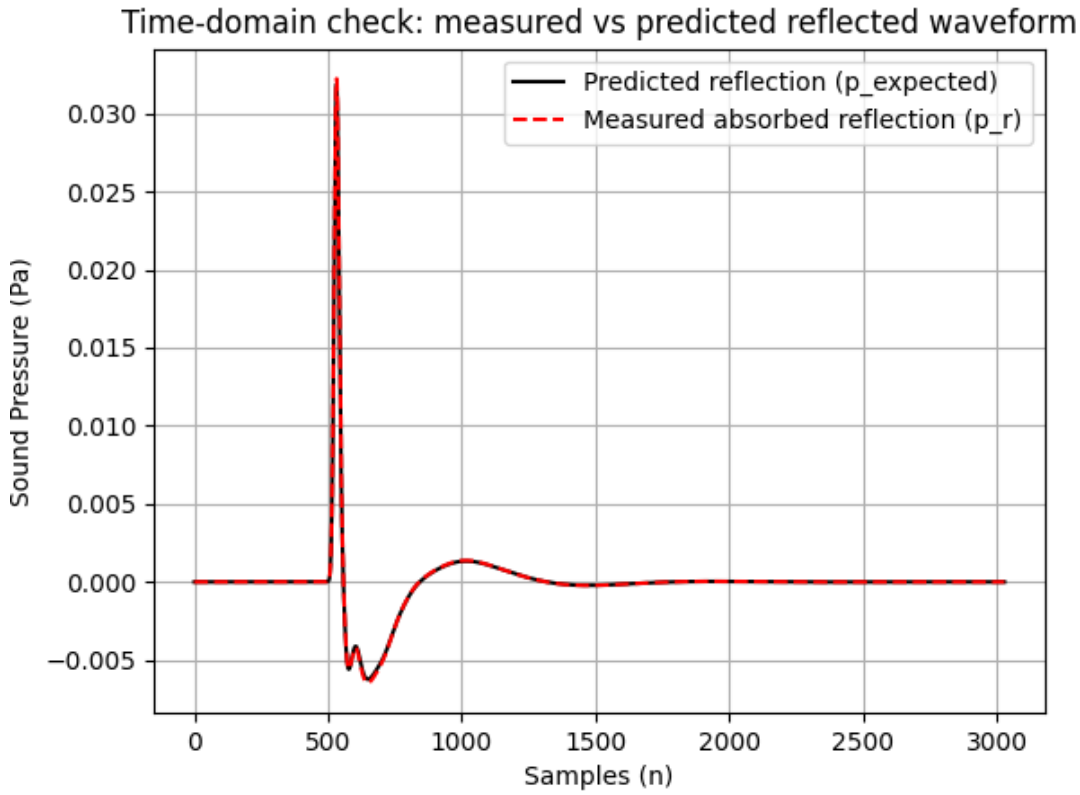


Figure C.2: Comparison of the numerical and expected sound pressure reflected at a receiver near a wall with impedance matching the one from the reflection in Figure C.1.

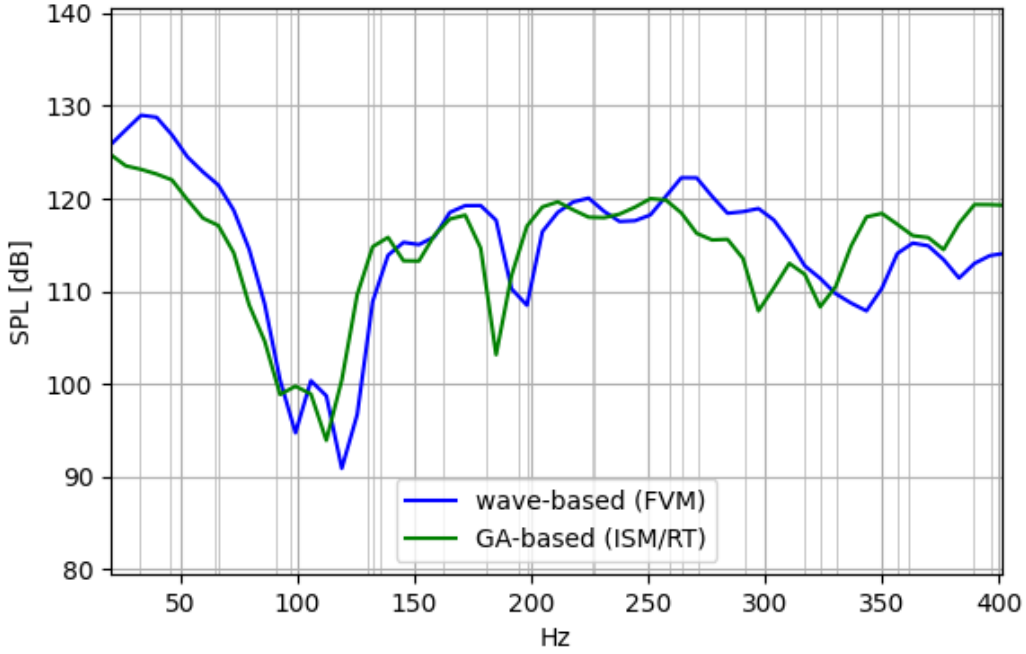


Figure C.3: Comparison of pure wave-based and pure GA-based simulation at a receiver in a shoebox-shaped empty domain with acoustic absorption of 0.5 on all surfaces, similar as in the thesis for Wayverb [8].

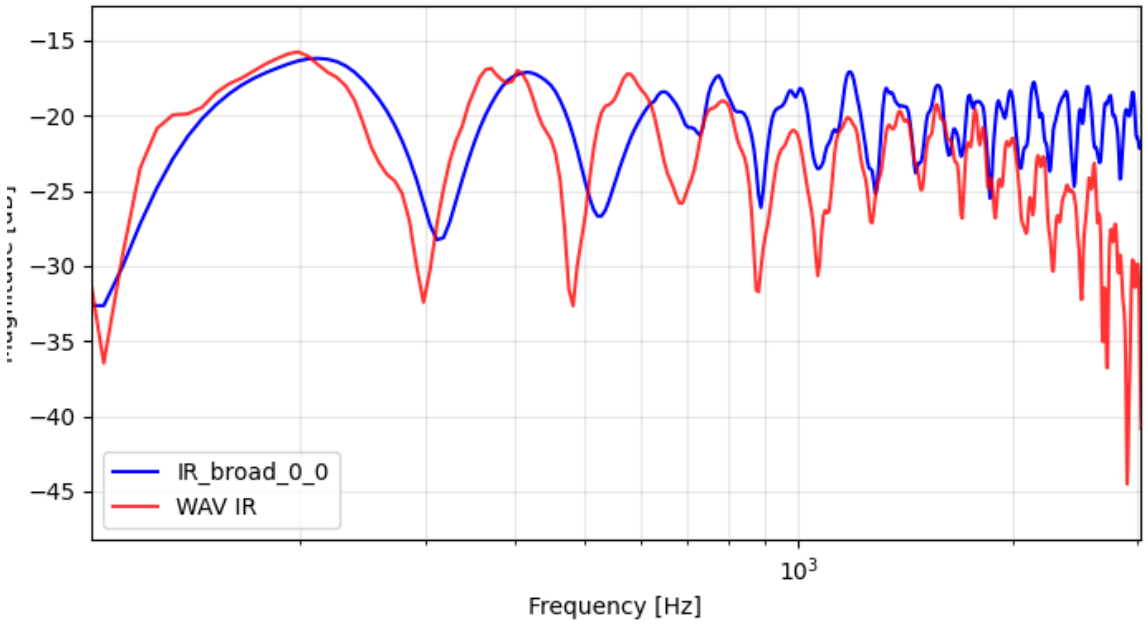


Figure C.4: Comparison of spectrum of F.A.S.E.S (in blue) at scene RS1 of BRAS (the Benchmark for Room Acoustical Simulation) with its dataset [9] (in blue), absorbing case, for source LS02 and receiver MP02, cutoff at 700 Hz.

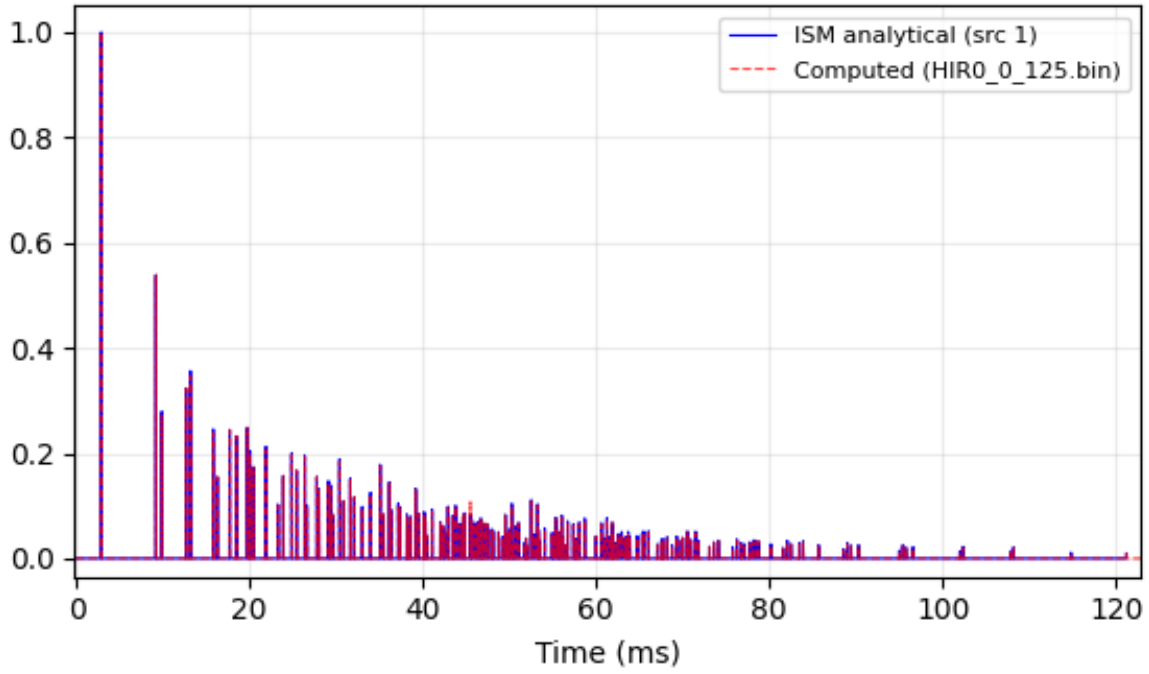


Figure C.5: Match between a F.A.S.E.S simulated shoebox-shaped room impulse response and an analytical response for the same case.

Bibliography

[1] R. J. Leveque, *Finite-Volume Methods for Hyperbolic Problems*. Cambridge University Press, 2002.

[2] L. Kinsler, A. Frey, A. Coppens, and J. Sanders, *Fundamentals of Acoustics*. Wiley, 2000.

[3] H. Wang and M. Hornikx, "Time-domain impedance boundary condition modeling with the discontinuous Galerkin method for room acoustics simulations," *The Journal of the Acoustical Society of America*, vol. 147, no. 4, pp. 2534–2546, 2020.

[4] J. Sheaffer, M. Van Walstijn, and B. Fazenda, "A physically-constrained source model for fdtd acoustic simulation," 15th *International Conference on Digital Audio Effects*, 2012.

[5] J. Allen and D. Berkley, "Image method for efficiently simulating small-room acoustics," *The Journal of the Acoustical Society of America*, 1979.

[6] H. Autio, N.-G. Vardaxis, and D. Bard Hagberg, "The influence of different scattering algorithms on room acoustic simulations in rectangular rooms," *Buildings*, vol. 11, no. 9, p. 414, 2021.

[7] R. Heinz, "Binaural room simulation based on an image source model with addition of statistical methods to include the diffuse sound scattering of walls and to predict the reverberant tail," *Applied Acoustics*, vol. 38, no. 2–4, pp. 145–159, 1993.

[8] M. R. Thomas, "Wayverb: A graphical tool for hybrid room acoustics simulation," Master's thesis, University of Huddersfield, 2017.

[9] L. Aspöck, F. Brinkmann, D. Ackermann, S. Weinzierl, and M. Vorländer, "BRAS – Benchmark for room acoustical simulation," 2020. Dataset, CC BY-SA 4.0.